Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If $0 < \theta < \pi/2$, and $f(\theta) = (1 + \cot\theta - \operatorname{cosec}\theta)(1 + \tan\theta + \sec\theta)$, then $f(\pi/4)$ is equal to

(A) 0

(B) 1

(C) 2

(D) $2 + \sqrt{2}$

Q.2

Let $f(\theta) = (1 + \cos\theta)(1 + \cos 3\theta)(1 + \cos 5\theta)(1 + \cos 7\theta)$ then $f\left(-\frac{\pi}{8}\right)$ is equal to:

(A) $-1/8$

(B) $1/4$

(C) $-1/4$

(D) $1/8$

Q.3

The acute angle of a rhombus whose side is a mean proportional between its diagonals is

(A) 15°

(B) 20°

(C) 30°

(D) 80°

Q.4

If $\tan p\theta = \tan q\theta$, then the values of θ form an A.P. with common difference

(A) $\pi/(p + q)$

(B) π/p

(C) π/q

(D) $\pi/(p - q)$

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Q.5

$\cos(540^\circ - \theta) - \sin(630^\circ - \theta)$ is equal to

(A) 0

(B) $2\cos\theta$

(C) $2\sin\theta$

(D) $\sin\theta - \cos\theta$

Q.6

The least positive root of the function $\sin x - \pi/2 + 1 = 0$ lies in the interval

(A) $[0, \pi/2]$

(B) $[\pi/2, \pi]$

(C) $[\pi/2, 3\pi/2]$

(D) none of these

Q.7

The value of $\tan^2\theta \sec^2\theta \left(\cot^2\theta - \cos^2\theta\right)$ is

(A) 0

(B) 1

(C) -1

(D) $\frac{1}{2}$

Q.8

If $0 \leq x \leq \pi$ and $81^{\sin^2 x} + 81^{\cos^2 x} = 30$, then $x =$

(A) $\pi/6$

(B) $\pi/2$

(C) $\pi/4$

(D) $3\pi/4$

Q.9

If $(\sec\alpha + \tan\alpha)(\sec\beta + \tan\beta)(\sec\gamma + \tan\gamma)$
$= \tan\alpha\tan\beta\tan\gamma$, then expression
$(\sec\alpha - \tan\alpha)(\sec\beta - \tan\beta)(\sec\gamma - \tan\gamma)$
is equal to

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(A) $\cot\alpha\cot\beta\cot\gamma$

(B) $\tan\alpha\tan\beta\tan\gamma$

(C) $\cot\alpha+\cot\beta+\cot\gamma$

(D) $\tan\alpha+\tan\beta+\tan\gamma$

Q.10

If $m=\operatorname{cosec}\theta-\sin\theta$ and $n=\sec\theta-\cos\theta$, then $m^{2/3}+n^{2/3}=$

(A) $(mn)^{-2/3}$

(B) $(mn)^{2/3}$

(C) $(mn)^{-1/3}$

(D) $(mn)^{1/3}$

Q.11

If $(1+\sin\alpha)(1+\sin\beta)(1+\sin\gamma)=(1-\sin\alpha)\,(1-\sin\beta)(1-\sin\gamma)=k$, then k is equal to :

(A) $2\cos\alpha\cos\beta\cos\gamma$

(B) $-\cos\alpha\cos\beta\cos\gamma$

(C) $\sin\alpha\cos\beta\cos\gamma$

(D) $2\sin\alpha\sin\beta\sin\gamma$

Q.12

If $p_n=\cos^n\theta+\sin^n\theta$, then p_n-p_{n-2} $=kp_{n-4}$, where :

(A) $\mathrm{k}=1$

(B) $\mathrm{k}=-\sin^2\theta\cos^2\theta$

(C) $\mathrm{k}=\sin^2\theta$

(D) $\mathrm{k}=\cos^2\theta$

Q.13

Find the value of $\cot 5^\circ\cot 10^\circ\ldots\ldots\cot 85^\circ$

(A) 1

(B) -1

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(C) 2

(D) -2

Q.14

Value of $2\left(\sin^6\theta+\cos^6\theta\right)$
$-3\left(\sin^4\theta+\cos^4\theta\right)+1$ is

(A) 2

(B) 0

(C) 4

(D) 6

Q.15

If $x\cos\theta+y\sin\theta=z$, then what is the value of $(x\sin\theta-y\cos\theta)^2$?

(A) $x^2+y^2-z^2$

(B) $x^2-y^2-z^2$

(C) $x^2-y^2+z^2$

(D) $x^2+y^2+z^2$

Q.16

$(1-\sin A+\cos A)^2$ is equal to

(A) $2(1-\cos A)(1+\sin A)$

(B) $2(1-\sin A)(1+\cos A)$

(C) $2(1-\cos A)(1-\sin A)$

(D) None of the above

Q.17

What is $\sin^2(3\pi)+\cos^2(4\pi)+\tan^2(5\pi)$ equal to ?

(A) 0

(B) 1

(C) 2

(D) 3

Q.18

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What is $\frac{1-\tan 2^0 \cot 62^\circ}{\tan 152^\circ - \cot 88^0}$ equal to?

(A) $\sqrt{3}$

(B) $-\sqrt{3}$

(C) $\sqrt{2}-1$

(D) $1-\sqrt{2}$

Q.19

The difference of two angles is 1°; the circular measure of their sum is 1 . What is the smaller angle in circular measure ?

(A) $\left[\frac{180}{\pi}-1\right]$

(B) $\left[1-\frac{\pi}{180}\right]$

(C) $\frac{1}{2}\left[1-\frac{\pi}{180}\right]$

(D) $\frac{1}{2}\left[\frac{180}{\pi}-1\right]$

Q.20

What is the angle(in circular measure) between the hour hand and the minute hand of a clock when the time is half past 4 ?

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$

(D) None of these

Answers & Solutions

Q.1 Answer:

2

Solution:

$$f(\theta)=\left(1+\frac{\cos\theta}{\sin\theta}-\frac{1}{\sin\theta}\right)\left(1+\frac{\sin\theta}{\cos\theta}+\frac{1}{\cos\theta}\right)$$
$$=\left(\frac{\sin\theta+\cos\theta-1}{\sin\theta}\right)\left(\frac{\sin\theta+\cos\theta+1}{\cos\theta}\right)$$
$$=\frac{(\sin\theta+\cos\theta)^2-1}{\sin\theta\cos\theta}=\frac{1+2\sin\theta+\cos\theta-1}{\sin\theta\cos\theta}=2$$
$$\therefore \quad f(\pi/4)=2$$

Q.2 Answer:

$1/8$

Solution:

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As $f(-\theta) = f(\theta)$, we evaluate $f(\pi/8)$. For $\theta = \pi/8, 3\theta = \pi/2 - \pi/8, 5\theta = \pi/2 + \pi/8$ and $7\theta = \pi - \pi/8$, so that

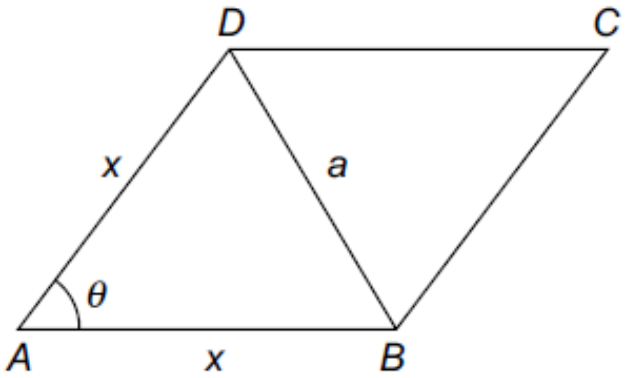
$$
\begin{aligned}
f\left(\frac{\pi}{8}\right) &= \left(1+\cos\frac{\pi}{8}\right)\left(1+\sin\frac{\pi}{8}\right)\left(1-\sin\frac{\pi}{8}\right)\left(1-\cos\frac{\pi}{8}\right) \\
&= \cos^2\left(\frac{\pi}{8}\right)\sin^2\left(\frac{\pi}{8}\right) \\
&= \frac{1}{4}\left[2\sin\left(\frac{\pi}{8}\right)\cos\left(\frac{\pi}{8}\right)\right]^2 \\
&= \frac{1}{4}\sin^2\left(\frac{\pi}{4}\right) = \frac{1}{8}
\end{aligned}
$$

Q.3 Answer:

30°

Solution:

Let the length of the diagonals of the rhombus be a and b and x be its side then



$\frac{x}{a} = \frac{b}{x} \Rightarrow x^2 = ab$

Now $\cos\theta = \frac{x^2+x^2-a^2}{2x\cdot x} = \frac{2ab-a^2}{2ab} = 1 - \frac{a}{2b}$

Similarly $\cos(180^\circ - \theta) = 1 - \frac{b}{2a}$

$\Rightarrow \frac{a}{2b} = 1 - \cos\theta, \frac{b}{2a} = 1 + \cos\theta$

$\Rightarrow 1 - \cos^2\theta = \frac{1}{4} \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = 30^\circ$

Q.4 Answer:

$\pi/(p-q)$

Solution:

$p\theta = n\pi + q\theta \Rightarrow (p-q)\theta = n\pi \Rightarrow \theta = \frac{n\pi}{p-q}$.

So the values of θ form an A.P. with common difference $\pi/(p-q)$.

Q.5 Answer:

0

Solution:

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$\cos(540^\circ - \theta) = \cos(6(\pi/2) - \theta) = -\cos\theta$
$\sin(630^\circ - \theta) = \sin(7(\pi/2) - \theta) = -\cos\theta$

Q.6 Answer:

$[0, \pi/2]$

Solution:

$\sin x = \frac{\pi}{2} - 1 = \frac{22}{14} - 1 = \frac{8}{14}$
$= \frac{4}{7}$ **so the least positive root lies in** $[0, \pi/2]$

Q.7 Answer:

1

Solution:

$\tan^2\theta \sec^2\theta \left(\cot^2\theta - \cos^2\theta\right)$
$= \sec^2\theta \left(\tan^2\theta \cot^2\theta - \tan^2\theta \cos^2\theta\right)$
$= \sec^2\theta \left(1 - \frac{\sin^2\theta}{\cos^2\theta}\cos^2\theta\right) = \sec^2\theta \left(1 - \sin^2\theta\right)$
$= \sec^2\theta \cdot \cos^2\theta = 1$

Q.8 Answer:

$\pi/6$

Solution:

We have, $81^{\sin^2 x} + 81^{\cos^2 x} = 30$
Now check by options, put $x = \frac{\pi}{6}$
then $(81)^{\sin^2 \pi/6} + (81)^{\cos^2 \pi/6} = 30$
$\Rightarrow (81)^{1/4} + (81)^{3/4} = 30 \Rightarrow 30 = 30$

Q.9 Answer:

$\cot\alpha \cot\beta \cot\gamma$

Solution:

$(\sec\alpha + \tan\alpha)(\sec\beta + \tan\beta)(\sec\gamma + \tan\gamma)$
$= \tan\alpha \tan\beta \tan\gamma$
$\Rightarrow \left(\sec^2\alpha - \tan^2\alpha\right)\left(\sec^2\beta - \tan^2\beta\right)\left(\sec^2\gamma - \tan^2\gamma\right)$
$= \tan\alpha \tan\beta \tan\gamma (\sec\alpha - \tan\alpha)(\sec\beta - \tan\beta)$
$(\sec\gamma - \tan\gamma)$
$\Rightarrow \quad (\sec\alpha - \tan\alpha)(\sec\beta - \tan\beta)(\sec\gamma - \tan\gamma)$
$= \cot\alpha \cot\beta \cot\gamma$

Q.10 Answer:

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$(mn)^{-2/3}$

Solution:

We have, $mn = (\operatorname{cosec}\theta - \sin\theta)(\sec\theta - \cos\theta)$

$= \left(\frac{1}{\sin\theta} - \sin\theta\right)\left(\frac{1}{\cos\theta} - \cos\theta\right)$

$= \frac{1-\sin^2\theta}{\sin\theta} \times \frac{1-\cos^2\theta}{\cos\theta}$

$= \frac{\cos^2\theta}{\sin\theta} \times \frac{\sin^2\theta}{\cos\theta} = \sin\theta \cdot \cos\theta$

$\therefore m^{2/3} + n^{2/3} = \left(\frac{\cos^2\theta}{\sin\theta}\right)^{2/3} + \left(\frac{\sin^2\theta}{\cos\theta}\right)^{2/3}$

$= \frac{\cos^{4/3}\theta}{\sin^{2/3}\theta} + \frac{\sin^{4/3}\theta}{\cos^{2/3}\theta} = \frac{\cos^2\theta+\sin^2\theta}{(\sin\theta\cdot\cos\theta)^{2/3}}$

$= \frac{1}{(mn)^{2/3}} = (mn)^{-2/3}$

Q.11 Answer:

$-\cos\alpha\cos\beta\cos\gamma$

Solution:

$\text{If}(1+\sin\alpha)(1+\sin\beta)(1+\sin\gamma) = k$

And $(1-\sin\alpha)(1-\sin\beta)(1-\sin\gamma) = k$

The the value of $k^2 = k \cdot k = (1+\sin\alpha)(1+\sin\beta)(1+\sin\gamma)(1-\sin\alpha)$

$(1-\sin\beta)(1-\sin\gamma)$

$= (1+\sin\alpha)(1-\sin\alpha)(1+\sin\beta)(1-\sin\beta)$

$(1+\sin\gamma)(1-\sin\gamma)$

$= \left(1-\sin^2\alpha\right)\left(1-\sin^2\beta\right)\left(1-\sin^2\gamma\right)$

$\Rightarrow k^2 = \cos^2\alpha\cos^2\beta\cos^2\gamma$

$\therefore \quad k = +-\cos\alpha\cos\beta\cos\gamma$

Q.12 Answer:

$\mathrm{k} = -\sin^2\theta\cos^2\theta$

Solution:

$p_n - p_{n-2} = \left(\cos^n\theta + \sin^n\theta\right) - \left(\cos^{n-2}\theta + \sin^{n-2}\theta\right)$

$= \cos^{n-2}\theta\left(\cos^2\theta - 1\right) + \sin^{n-2}\theta\left(\sin^2\theta - 1\right)$

$= -\sin^2\theta\cos^{n-2}\theta - \cos^2\theta\sin^{n-2}\theta$

$= -\sin^2\theta\cos^2\theta\left(\cos^{n-4}\theta + \sin^{n-4}\theta\right)$

$= -\sin^2\theta\cos^2\theta p_{n-4} = \mathrm{k}p_{n-4}$

$\Rightarrow k = -\sin^2\theta\cos^2\theta$

Q.13 Answer:

1

Solution:

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$$\cot 5^\circ \cot 10^\circ \ldots\ldots\ldots \cot 85^\circ$$
$$= \cot 5^\circ \cot 10^\circ \ldots \cot (90^\circ - 10^\circ) \cot (90^\circ - 5^\circ)$$
$$= \cot 5^\circ \cot 10^\circ \ldots\ldots \tan 10^\circ \tan 5^\circ$$
$$= (\tan 5^\circ \cot 5^\circ)(\tan 10^\circ \cot 10^\circ) \ldots\ldots$$
$$= (1)(1)(1) \ldots\ldots = 1$$

Q.14 Answer:

0

Solution:

We have

$$2\left(\sin^6\theta + \cos^6\theta\right) - 3\left(\sin^4\theta + \cos^4\theta\right) + 1$$
$$= 2\left[\left(\sin^2\theta\right)^3 + \left(\cos^2\theta\right)^3\right] - 3\left(\sin^4\theta + \cos^4\theta\right) + 1$$
$$= 2\left[\left(\sin^2\theta + \cos^2\theta\right)\left(\sin^4\theta + \cos^4\theta - \sin^2\theta\ \cos^2\theta\right)\right] - 3\left(\sin^4\theta + \cos^4\theta\right) + 1$$
$$= \left[2\sin^4\theta + 2\cos^4\theta - 2\sin^2\theta\cos^2\theta\right] - 3\sin^4\theta - 3\cos^4\theta + 1$$
$$= -\left(\sin^2\theta + \cos^2\theta\right)^2 + 1 = -1^2 + 1 = -1 + 1 = 0$$

Q.15 Answer:

$x^2 + y^2 - z^2$

Solution:

Here, $z = x\cos\theta + y\sin\theta$

$$z^2 = x^2\cos^2\theta + y^2\sin^2\theta + 2xy\sin\theta\cos\theta$$
$$\Rightarrow 2xy\sin\theta\cos\theta = z^2 - x^2\cos^2\theta - y^2\sin^2\theta$$
$$\text{Let, } L = (x\sin\theta - y\cos\theta)^2$$
$$\Rightarrow L = x^2\sin^2\theta + y^2\cos^2\theta - 2xy\sin\theta\cos\theta$$
$$\Rightarrow L = x^2\sin^2\theta + y^2\cos^2\theta - \left[z^2 - x^2\cos^2\theta - y^2\sin^2\theta\right]$$
$$\Rightarrow L = x^2\left[\sin^2\theta + \cos^2\theta\right] + y^2\left[\sin^2\theta + \cos^2\theta\right] - z^2$$
$$\Rightarrow L = x^2 + y^2 - z^2$$

Q.16 Answer:

$2(1 - \sin A)(1 + \cos A)$

Solution:

$$(1 - \sin A + \cos A)^2$$
$$= 1 + \sin^2 A + \cos^2 A - 2\sin A$$
$$-2\sin A \cdot \cos A + 2\cos A$$
$$= 2 - 2\sin A - 2\sin A\cos A + 2\cos A$$
$$= 2(1 - \sin A) + 2\cos A(1 - \sin A)$$
$$= 2(1 + \cos A)(1 - \sin A)$$

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Q.17 Answer:

1

Solution:

$\sin^2(3\pi) + \cos^2(4\pi) + \tan^2(5\pi)$
$= \sin^2(3\pi) + \cos^2(\pi + 3\pi) + \tan^2(5\pi)$
$= \left(\sin^2(3\pi) + \cos^2(3\pi)\right) + \tan^2(2 \times 2\pi + \pi)$
$= 1 + \tan^2\pi = \sec^2\pi = 1$

Q.18 Answer:

$-\sqrt{3}$

Solution:

$L = \frac{1-\tan 2^\circ \cot 62^\circ}{\tan 152^\circ - \cot 88^\circ} = \frac{1-\tan 2^\circ \cot(90-28)^\circ}{\tan(180-28)^\circ - \cot(90-2)^\circ}$
$\Rightarrow L = \frac{1-\tan 2^\circ \tan 28^\circ}{-\tan 28^\circ - \tan 2^\circ} = -\left[\frac{1-\tan 2^\circ \tan 28^\circ}{\tan 2^\circ + \tan 28^\circ}\right]$
$\Rightarrow L = -\frac{1}{\tan(2+28)^\circ} = -\frac{1}{\tan 30^\circ} = -\sqrt{3}$
$\left[\because \tan(A+B) = \frac{\tan A + \tan B}{1-\tan A \tan B}\right.$

Q.19 Answer:

$\frac{1}{2}\left[1 - \frac{\pi}{180}\right]$

Solution:

Let the angles are α and β, then $\alpha - \beta = 1^\circ$
$\Rightarrow \quad \alpha - \beta = \frac{\pi}{180^\circ}$ **is circular measure . . . (i)**
As given, $\alpha + \beta = 1$ **On solving Eqs. (i) and (ii), we get,** $\alpha = \frac{1}{2}\left[1 + \frac{\pi}{180^\circ}\right]$ **and** $\beta = \frac{1}{2}\left[1 - \frac{\pi}{180^\circ}\right]$
B is the smaller angle. Hence, smaller angle $= \frac{1}{2}\left[1 - \frac{\pi}{180^\circ}\right]$

Q.20 Answer:

$\frac{\pi}{4}$

Solution:

Angle traced by the hour hand in 12 hours $= 360^\circ$
Angle traced byit in $4\text{hr}30\,\text{min}\left(4\text{ h} + \frac{30}{60}\text{hr}\right) = \frac{9}{2}\text{hr}$
$= \frac{9}{2} \times \frac{360}{12} = 135^\circ$
Angle traced by minute hand is 60° min $= 360^\circ$ **Angle traced by it in** 30 min $= \frac{30}{60} \times 360 = 180^\circ$
Required Angle $= 180^\circ - 135^\circ = 45^\circ$
$\Rightarrow 45 \times \frac{\pi}{100} = \frac{\pi}{4}$ **radian**

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